

A Packing of 8 Four-Spaces in $\mathbb{R}^{13}$

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October 9, 2026

Abstract

We give 8 4-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 3.164606. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 3.115123, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 8 subspaces of dimension 4 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 3.164606\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 4$. The cell $N = 8$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 3.115123, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(13 - 4)/13 \cdot 8/7 = 3.164835$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n4N8.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
527434403942525013948244273057869267238685269591901620467085363674313464
269289657657133435458172001305663409946169829129503556479337910429654801
370355590826167950260019515009191040940098739339
```

denominator:

```
166666666666841231620475148135754371899159603517745511569638448454831994
123860850837468882273727223930764051334131392355703062926137473281016107
150829362656764391021655541333957720948619579485
```

B Coordinates

8 blocks of 13 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m13n4N8.txt`.

```
-178859567228 293369859558 4933740310 -38970072467395813535161 115344759163 313660075955 -200521816942166406375353 -357221805559 -79938238399 -146398066290
627619933930 475228149073 -16882305217 -8816890002427894730052 68301776175 -401739008848 -6098902722187800852253 -346821684182 394810858768 -43745833589
-90918551882 428974453384 615480980876 -2000346815244723537191 -14463909939 -280822218689 1655186720737676343321 -277872394583 63383705143 393063076921
126117440054 20593370257 365165124402 38157282619 -103905667893 -132390323579 48259607961 -212968037847160028003 498524957414 -229955623481 456642603445
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```